Pole Positions from the Mamainse Point Lavas and their Bearing on a Keweenawan Pole Path and Polarity Sequence[1]

W. A. ROBERTSON

Earth Physics Branch, Department of Energy, Mines, and Resources, Ottawa, Canada

Received May 2, 1973

Revision accepted for publication July 5, 1973

Thermal and alternating field studies of the remanent magnetization of the Mamainse Point lavas, on the east shore of Lake Superior, indicate stable magnetization. The flows form two groups of normally magnetized (N) flows, each overlying reversely magnetized (R) flows. It is suggested that the sequence consists of a single sequence of R flows overlain by N flows, the sequence being repeated by faulting. A mean pole position calculated from 15 N sites is 30° N, 183° E, (dp = 5, dm = 8), and the pole position from 12 R sites is 49° N, 232° E (dp = 14, dm = 16). Reasons are given for believing that this difference in pole position between the older (R) and younger (N) flows was caused by relative movement of the Keweenawan area and the rotation axis of the earth, and not to secondary components of magnetization acquired since the rocks cooled. Review of the Keweenawan paleomagnetic data reveals a consistent picture of an older reversely magnetized and younger normally magnetized magnetic field (if allowance is made for radiogenic errors). The time at which the field switched from reversed to normal polarity may provide a valuable time datum traceable in the rocks from continent to continent.

Des études thermales et de champ alternatif du magnétisme résiduel des laves de "Mamainse Point" sur la rive orientale du lac Supérieur indiquent que le magnétisme est stable. Ces laves forment deux groupes de coulées de magnétisme (N) normal chacune recouvrant des coulées de magnétisme (R) inversé.

Nous proposons que cette suite soit considérée comme une seule de coulées R recouverte de coulées N, suites qui se répète à cause de failles.

Une attitude moyenne du pôle, calculée à partir de quinze localités N, est 30° N, 183° E, (dp = 5, dm = 8), et une attitude moyenne du pôle, calculée à partir de douze localités R, est 49° N, 232° E (dp = 14, dm = 16). Nous donnons les raisons qui font croire que cette différence dans l'attitude du pôle entre les laves plus anciennes (R) et les plus jeunes (N) a été causée par un mouvement relatif de la région Keweenavienne et de l'axe de rotation de la terre, et non pas par des composantes secondaires du magnétisme depuis le refroidissement des roches. Une revision des données paléomagnétiques du Keweenavien laisse voir une persistance d'un champ magnétique plus vieux inversé et d'un champ plus jeune et normal (avec tolérance pour erreur radiogéniques).

Le moment où le champ terrestre a changé de la polarité inverse à la polarité normale peut établir un point de repère valable dans l'étude des roches d'un continent à un autre.

[Traduit par le journal]

Introduction

DuBois (1962) showed that the magnetization directions from the Mamainse Point lavas were of opposite sense and also gave a different pole position from those at Alona Bay. Recently Palmer (1970) sampled the Mamainse Point lavas in more detail and showed that the well-exposed section on the north side of Mamainse Point contains two normally magnetized zones stratigraphically above each of two zones of reversed polarity, which yielded pole positions similar to that from the lavas at Alona Bay.

Recent paleomagnetic work on Keweenawan rocks around Lake Superior, by Vincenz (1968) and Books (1968) on the Portage Lake lavas, by Beck (1970) on the Minnesota intrusives, by Palmer (1970) on the North Shore volcanics, Gargantua Head lavas, and the Osler lavas, and by Robertson and Fahrig (1971) on the Thunder Bay dikes and Logan sills, has yielded a consistent picture in the Keweenawan of an older period of reversed magnetization, followed by a younger normal period, the latter yielding a pole position about 30°W of the pole from the reversely magnetized rocks.

The double reversal reported by Palmer (1970) in the Mamainse Point lavas is a glaring inconsistency in this Canadian Shield Neohelikian pattern. This paper reports detailed laboratory work on the lavas and intrusive rocks that crop out in the Mamainse Point

[1]Contribution from the Earth Physics Branch, No. 458.

area. The aim is to test, firstly, whether the double reversal is caused by three reversals of the earth's magnetic field, or by faulted repetition of a single reversal and secondly, whether the departure from a 180° reversal noted above is caused by partial instability as suggested by Palmer (1970), or by a genuine polar shift relative to the Keweenawan flows between the times the two groups of rocks acquired their magnetization (Robertson and Fahrig 1971).

Geology and Age

The lavas at Mamainse Point, Gargantua Point, and Alona Bay rest with angular unconformity on Archean granitic rocks, metasediments, and metavolcanic rocks (Annells 1973). They form the eastern edge of a large trough and dip towards the center of the trough. They are similar to flood basalts elsewhere, such as the Coppermine River, Columbia River, and Icelandic basalts (White 1960).

The section at Mamainse Point appears to have a total thickness approaching 6000 m (~20 000 ft) (Fig. 1). It comprises over 300 flows with interbedded polymictic conglomerates and coarse red sandstones. The sediments make up 24% of the total thickness of the section (Annells 1973). The red color of the sediments suggests an arid environment (Schwartzback 1963) of the piedmont-valley flat type (Van Houton 1961). The igneous rocks are strongly bimodal, the modes being basaltic and rhyolitic. Much of the acid mode appears to be intrusive, but the basic mode occurs mainly as basaltic lava flows. Many of the flows are 1.5–9.0 m (5–30 ft) thick, but they range in thickness up to 30 m (100 ft).

A remarkable 21-m (70-ft) marker flow, known as the "Daisy Stone," because of large spherulitic plagioclase clusters (Giblin 1969; Annells 1973), is clearly visible near the base of the flows, both at Mamainse Point and Gargantua Point. This shows that these separate areas of basaltic lavas were part of the same sequence, a fact confirmed by their polarity and pole position (*see* below).

The lower flows in particular display mild metamorphic effects. The olivines are invariably pseudomorphed by saponite and hematite (Annells 1973), and the opaque minerals are commonly oxidized to hematite, with the titanium-rich types going over to leucoxene and sphene. Epidote is also common in the lower flows. Some of these alterations may be deuteric, but some may be due to the effect of burial.

K/Ar ages from the Mamainse Point lavas, which should be regarded as minimum because of the probability of argon loss, range in age from 800 to 1055 m.y. (Wanless *et al.* 1967, 1968). Recently Van Schmus (1971) has obtained a Rb/Sr isochron of 1070 ± 50 m.y. (based on a λ (^{87}Rb) = 1.39×10^{-11} yr^{-1}) from 6 rhyolites within the sequence. Thus the Mamainse Point (and Gargantua Point) lavas are in the same age bracket as the Duluth Gabbro (Faure *et al.* 1969; Silver and Green 1963) and the North Shore volcanics, and probably also as the Portage Lake lavas, the Osler lavas, and the Coldwell complex, although the radiogenic age for the Duluth Gabbro (1115 m.y. Faure *et al.* (1969)) extends the upper limit.

Results

Thermal and alternating field demagnetization of a pair of specimens from each site, using the apparatus described by Roy *et al.* (1971*a*) and Roy *et al.* (1971*b*), yielded the results of Table 1, which are divided into groups according to polarity. The four sites of intrusive rocks of Table 2 were an acid intrusion and a lava baked by it, and acid and basic phases of a multiple intrusion (Annells 1973). They all had normal polarity and intruded the upper group (N1) of normally magnetized lavas; they have been put in a separate group.

Table 1 shows that the precision for all groups except the N2 lavas improved with heating, the optimum cleaning temperature ranging from 100 °C for the normally magnetized groups, to 540 °C for the R2 lava group. However, directions for all groups except R2 were not greatly scattered after any heating up to 540 °C. This behavior, in which direction changes are small over a wide blocking temperature range, is consistent with the idea that the remanent magnetization is derived from a single dominant magnetization, acquired in the earth's magnetic field either at the time of, or soon after cooling, with a small randomly directed component superposed in the low blocking temperature range. In the case of R2 the randomly directed component of the NRM was much more significant; this scattering of

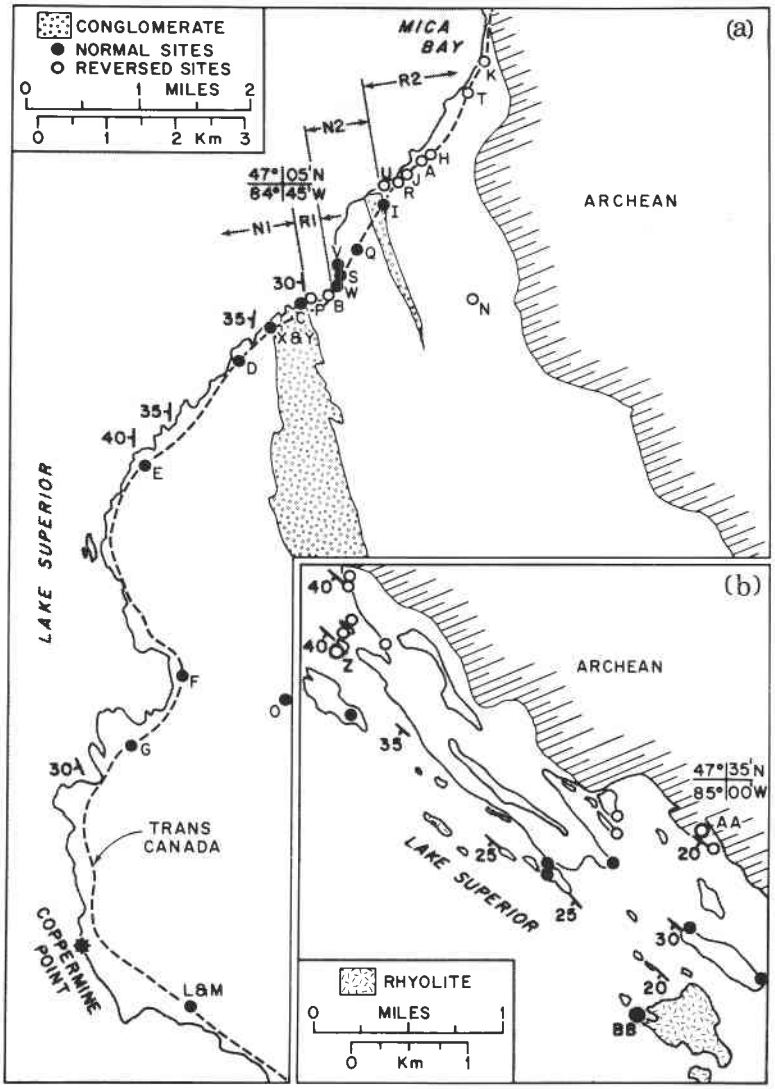


FIG. 1. (*a*) Geological sketch map of the Mamainse Point area (after Thomson 1953, and Giblin and Leahy 1965), showing the sampling sites. Thin conglomerates and the intrusive rocks have not been shown on the map. (*b*) Geological sketch map of the Gargantua Point area (after Ayres 1961), showing both new (lettered) and Palmer's (1970) sampling sites.

directions towards the bottom of the pile may be partly the result of deuteric metamorphism, the petrological effects of which were more visible in the lower part of the sequence (Annells 1973).

The thermal demagnetization curves shown in Fig. 2 are divided into the same groups as Tables 1 and 2. It is clear that all groups have a predominant blocking temperature range between 500 ° and 600 °C. Site F, which gives the earth's field direction, contains abnormally large components at the low temperature end of the blocking temperature spectrum. The increase after heating to 700 °C for sites P, I, and R is thought to be caused by chemical alteration during laboratory heating. Many of the lavas retained a notable component of magnetization after heating to 600 °C; this is above the blocking temperature of magnetite, but is to be expected in lavas containing hema-

TABLE 1. Pilot demagnetization

Alternating magnetic field

Peak fields in Oersteds	5 sites from Group N1				2 sites from Group R1				5 sites from Group N2				9 sites from Group R2				4 sites from intrusives			
	D	*I*	*k*	α	*D*	*I*	*k*	α	*D*	*I*	*k*	α	*D*	*I*	*k*	α	*D*	*I*	*k*	α
NRM	286	+34	58	10	106	−64	302	14	285	+40	37	13	128	−81	3.9	30	294	+34	57	12
50	294	+36	64	9.6	105	−65	479	11	*282*	*+39*	*54*	*11*	099	−82	5.2	25	*292*	*+31*	*88*	*9.8*
100	*295*	*+34*	*76*	*8.8*	103	−64	280	15	283	+39	43	12	110	−78	8.7	19	291	+33	45	14
150	294	+35	51	11	099	−64	317	14	282	+39	45	12	104	−77	12	15	290	+31	56	12
200	294	+36	55	10	*099*	*−62*	*1220*	*7.2*	283	+40	35	13	113	−77	17	13	292	+33	30	17
300	295	+36	53	11	100	−62	554	11	277	+43	20	18	*119*	*−76*	*27*	*10*	282	+34	9.5	31
400	296	+36	58	10	091	−57	219	17	282	+40	32	14	113	−76	20	12	277	+40	3.4	59
600	296	+35	44	12	090	−57	179	19	279	+44	33	14	116	−75	23	11	281	+41	2.2	83
1000	297	+35	42	12	090	−56	185	18	273	+38	19	18	117	−75	22	11	281	+38	2.0	91

Thermal

Temp. in °C	5 sites from Group N1				2 sites from Group R1				5 sites from Group N2				9 sites from Group R2				3 sites from intrusives			
	D	*I*	*k*	α	*D*	*I*	*k*	α	*D*	*I*	*k*	α	*D*	*I*	*k*	α	*D*	*I*	*k*	α
NRM	291	+40	31	14	104	−72	21	58	*291*	*+39*	*32*	*14*	125	−75	3.6	32	294	+34	31	23
100	*293*	*+38*	*57*	*10*	099	−68	38	42	286	+45	15	20	133	−80	4.3	28	*290*	*+31*	*34*	*21*
300	294	+36	38	13	097	−62	195	18	272	+46	23	16	109	−81	8.0	19	291	+32	27	24
400	295	+34	35	13	*094*	*−61*	*665*	*9.7*	268	+44	9.5	26	110	−80	11	16	287	+28	23	26
500	292	+34	19	18	096	−54	55	34	264	+47	8.3	28	107	−78	12	15	286	+30	20	29
540	293	+35	19	18	098	−53	62	32	261	+48	6.9	31	*104*	*−78*	*14*	*14*	280	+32	18	30
580	292	+35	19	18	091	−51	36	43	257	+50	5.4	36	098	−78	13	15	261	+36	7	51
620	292	+40	20	17	074	−27	1.4	—	250	+48	5.8	35	128	−80	6.4	22	271	+40	2.6	100
660	300	+36	7	31	104	−33	2.3	—	202	+47	2.3	66	114	−81	6.1	23	206	+27	1.7	—
700	286	+42	2.8	57	125	−07	0.8	—	208	+46	3.3	49	132	−84	3.0	36	166	+33	1.5	—

D is the declination east of true north. *I* is the inclination positive downwards.
k is Fisher's (1953) best estimate of precision, and α is the half angle of the cone of confidence at P = 0.95.

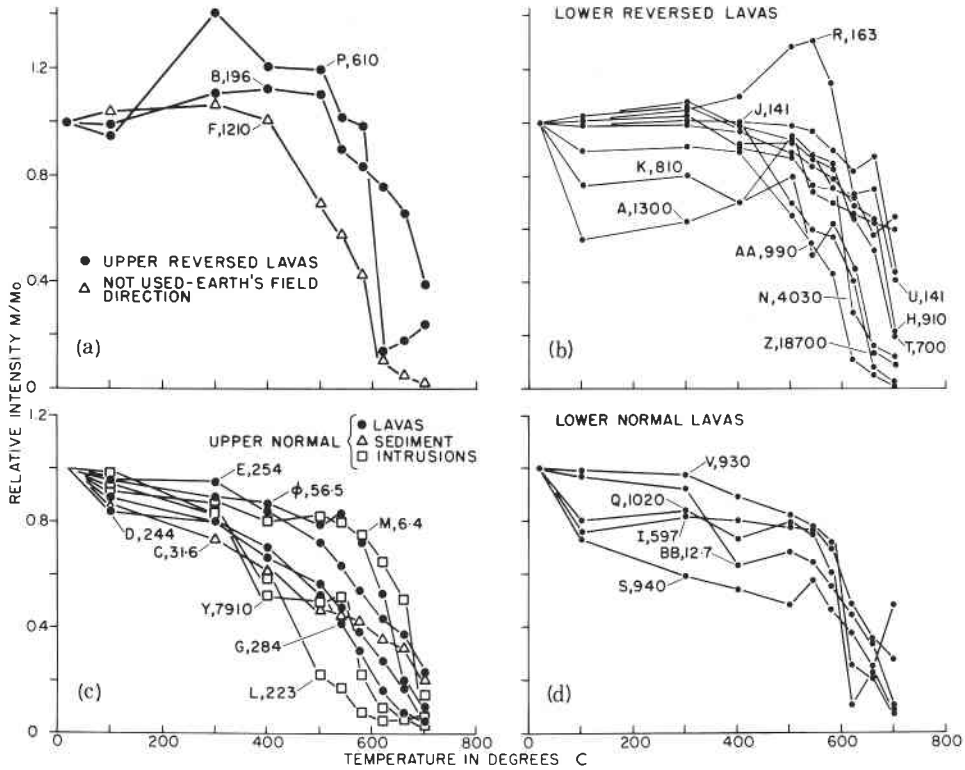


FIG. 2. Normalized thermal demagnetization curves for a specimen from each site. Letters refer to the sites of Fig. 1 and the numbers to the initial intensity of mangetization in e.m.u. $\times 10^{-6}$.

tite, as do many of the flows seen in both outcrop and polished sections. Since most of the magnetization has blocking temperatures higher than 500 °C, a temperature well in excess of that to which the rocks could have been raised during their post-cooling geological history, it is considered that the magnetization, and the direction associated with it, must have been acquired as the molten lava cooled. The time of initial cooling is also most likely to have been that at which the rubidium–strontium and potassium–argon clocks were started.

Comparison of the lower part of Table 1 with the upper part shows that the most precise results (those underlined) obtained by alternating magnetic-field cleaning were better grouped than the comparable most precise thermally cleaned groups. Hence for the final results, in order to obtain the greater accuracy, alternating field cleaning was used. The optimum peak field for cleaning the normal sites was 50–100 Oe and for the reversed sites 200–300 Oe, hence all specimens from normally magnetized sites were demagnetized in 100 Oe, and all specimens from reversely magnetized sites in 200 Oe.[2]

Comparison of Tables 1 and 3 indicates that the samples with high coercivity spectra contain predominant hematite, and those with low coercivity spectra contain magnetite as the major opaque constituent.

Experiments in progress on specimens from the Mamainse Point lavas (G. Pullaiah, personal communication) are designed to show the effect of elevated temperatures on the build up and dissipation of viscous magnetization.

Both the blocking temperature and coercivity

[2]The reason for this difference in optimum cleaning field between the normally and reversely magnetized rocks is not clear: one possibility is that the scattering component is not entirely random but has a residual closer in direction to that of the normally magnetized specimens (for instance the present field) and therefore has a greater proportional effect on the reversely magnetized specimens. The coercivity spectra of the pilot samples have a wide range in all groups, so that they overlap between the 4 lava groups.

TABLE 2. Site and group directions and pole positions

Site	*D*	*I*	*N*	*R*	A.M.	Pole Positions Lat.	Long.	dp	dm
Normal lavas cleaned in 100 A.F.									
Group N1 intrusives w.r.t. horizontal									
L	323	+49	5	4.922	830	57N	168E	9.4	14
M	302	+49	5	4.983	5.0	43N	185E	4.3	6.5
X	302	+51	5	4.950	11.8	44N	188E	7.9	12
Y	303	+59	5	4.979	3850	49N	197E	6.3	8.4
Group N1 intrusives w.r.t. beds									
L	295	+41	5	4.922	830	34N	185E	8.0	13
M	303	+21	5	4.983	5.0	30N	167E	2.7	5.2
X	286	+25	5	4.950	11.8	20N	183E	5.0	9.3
Y	282	+32	5	4.981	3850	21N	189E	3.4	6.0
Sites	292	+30	4	3.932	—	26N	181E	8.6	16
Group N1 lavas w.r.t. beds									
D	302	+43	5	4.991	220	40N	180E	2.8	4.6
E	299	+33	5	4.960	1700	33N	176E	4.9	8.7
G	299	+38	5	4.866	391	35N	180E	10	17
O	303	+32	5	4.849	342	35N	173E	9.5	17
Sites	301	+37	4	3.990	—	36N	178E	4.0	7.1
Group N1 sediments w.r.t. beds									
C	279	+30	5	4.949	31.2	18N	190E	5.4	9.6
Group N2 lavas w.r.t. beds									
I	268	+39	6	5.949	1490	14N	202E	4.8	8.1
Q	281	+45	5	4.947	304	26N	197E	7.1	11
S	308	+20	5	4.936	819	33N	163E	5.4	10
V	274	+40	4	3.969	799	20N	198E	6.9	11
W	316	+39	4	3.847	930	47N	165E	11	18
BB	293	+41	6	5.886	21.7	32N	186E	7.6	13
Sites	290	+39	6	5.760	—	30N	186E	10	18
Normal grand averages									
GPS	290	+34	4	3.955	—	29.6N	184.3E	7.4	13
STS	293	+35	15	14.603	—	29.5N	182.7E	4.4	7.6
SAM	293	+35	76	72.992	—	29.5N	182.8E	2.2	3.8

spectra of the intrusive rocks are in general lower than for the lavas. It is evident, as may be seen both in Table 2 and by a comparison of Figs. 2 and 3, that a sympathetic relationship exists between blocking temperature and coercivity as predicted by theory (Neel 1955; McElhinny 1973).[3]

Table 2 is a synopsis of the results obtained in the present study. Previous results obtained by DuBois (1962) and Palmer (1970) completely agree with regard to the polarity of the magnetization of the rocks (see Fig. 1b for Gargantua polarity), and all the lavas collected both by DuBois and by Palmer from Alona Bay were reversely magnetized.

Directions from the flows fall naturally into four groups based on their polarity (Palmer 1970). A thick succession of normally magnetized flows at the top (N1) is underlain by a thinner band (R1) that is reversely magnetized (*see* Fig. 1a). This is underlain by a thin band (N2) magnetized normally, which again rests on reversely magnetized lavas (R2).

The directions used in Table 2 are after correction for the geological tilt of the flows, so that they are considered to be relative to the

[3]In view of the high blocking temperatures and coercivities (of remanence) of some pilot samples it seems possible that stepwise thermal and alternating field cleaning may have a magnetic tempering effect, so that the initial magnetization is partly removed and partly 'hardened,' the latter part becoming more resistant to further treatment. If this is the case a single cleaning at the field or temperature best suited to the rock may be more effective in eliminating secondary components, than demagnetization in closely spaced stages, as in pilot runs.

TABLE 2. (*Concluded*)

Site	*D*	*I*	*N*	*R*	A.M.	Pole Positions Lat.	Long.	dp	dm
				Reversed lavas cleaned in 200 A.F.					
Group R1 lavas w.r.t. beds									
B	105	−55	5	4.978	513	34N	202E	5.7	8.0
P	090	−63	5	4.976	549	31N	219E	7.3	9.3
Sites	098	−59	2	1.991	—	33N	210E	26	36
Group R2 lavas w.r.t. beds									
A	289	−74	5	4.761	107	32N	209E	31	35
H	049	−79	6	5.601	559	32N	267E	36	37
J	069	−71	6	5.992	410	27N	238E	4.0	4.6
K	172	−72	8	7.278	302	79N	251E	26	25
N	101	−67	6	5.081	2060	40N	220E	44	53
R	125	−85	6	5.845	211	52N	261E	23	24
T	102	−76	5	4.997	768	46N	237E	3.3	3.6
U	103	−68	4	3.984	1450	42N	220E	9.4	11
Z	095	−65	6	5.940	10500	35N	219E	9.6	12
AA	133	−61	5	4.994	957	56N	194E	3.5	4.6
Sites	106	−77	10	9.676	—	48N	239E	16	17
Reversed grand averages									
GPS	100	−68	2	1.974	—	40.4N	221.3E	57	68
STS	104	−74	12	11.584	—	48.5N	232.0E	14	16
SAM	104	−74	67	62.218	—	46.5N	232.3E	8.0	8.8
F	322	+68	5	4.676	8300	65.3N	210E	32	38

D is the declination east of true north.
I is the inclination positive downwards.
N is the number of separately oriented samples.
R is the vector sum of the unit vectors (*N*) summed.
A.M. is the arithmetic mean intensity in e.m.u. $\times 10^{-6}$.
dp and dm are the semi-axes of the ovals of confidence of the pole positions at a probability P = 0.95.
GPS = Groups as units, STS = Sites as units, SAM = Samples as units.

horizontal at the time the lavas were poured out. Since they are basic flows their low viscosity at the time of extrusion would not sustain a significant initial dip. The intrusive rocks sampled were intruded into the top (N1) part of the sequence and all are normally magnetized. They are thought (Annells 1973) to be one phase of the same magmatic activity and are therefore essentially of the same age as the lavas they intrude. All those sampled (Table 2) are normally magnetized. The directions of the intrusive rocks were calculated both with respect to the present horizontal plane and also with respect to the bedding of nearby lava flows. The latter directions agree more closely with those of the lavas, from which it is concluded that the tilting took place after the emplacement of the intrusions.

Opaque Mineralogy

A synopsis of the examination of a polished section from each site is given in Table 3. Although the percentage of opaque minerals in the section and the proportion of hematite to magnetite differs widely from section to section, no trend is evident through the sequence. Sulfides appear to be absent in the lavas, but traces of native copper occur. Hematite is the predominant opaque mineral in most flows, and where this is the case the median destructive field is high, and generally the within-site dispersion is small (large k). The presence of ilmenite in most cases appears to extend the lower end of the blocking temperature range.

Flows in which hematite is the predominant opaque mineral have been separated from those in which magnetite predominates. The mean directions for both the hematite-rich and magnetite-rich flows (normally and reversely magnetized flows being treated separately) are shown in Fig. 4; it is evident that there is no significant difference between them. This is considered to be evidence favoring the idea of hematite formation very soon after the

TABLE 3. Opaque minerals and some magnetic properties
(The lavas are given in stratigraphic order, the youngest at the top)

Site	% Opaques	Color Index[1]	Fe_3O_4	Fe_2O_3	Other	Blocking temperature in 10 °C	m.d.f.[2]	Intensity × 10^6 e.m.u.	k	F
Intrusives										
L	1.2	A	++B r 20–300	—	−Ps tCu	Dist. 30–58	320	173	52	41
M	0.3	E	++sf 1–20	—	=Py	Dist. and 58–70	>1000	4.6	240	1.8
X	0.2	E	−C >300	++a<3	—	—	330	0.8	80	22
Y	1.5	A–B	++r 10–60 S150	—	=Su	30–40 and 58–62	140	7600	209	32
N1 lavas										
G	0.07	C–D	−s 1–60	=a 1–20	++I1c 1–500	Dist. 30–62	335	510	30	3.4
F[x]	1.0	C–E	−r 5–100	++a 1–400	+j 5–200	Dist. 40–62	500	1300	12	21
O	7.3	F	−r1 5–50	++1 & a 5–250	+I11 5–100	58–66	>1000	630	27	16
E	11.2	E	+1 30–400	++a & 1 30–500	=Ps	Dist. 30–70	>2000	2670	101	17
D	10.0	D	−r <50	++a & 1 60–600	−Ps	54–66	290	270	430	4.5
N1 sediment										
C	2.7		+af 1–40	+a 1–40	−Ps	Dist. 20–70	>1000	45	79	1.6
R1 lavas										
P	4.6	B–C	−s 1–10	++Ba 20–100 & 120–200	−I11 20–60	58–62	1460	640	169	3.6
B	9.0	E	=r 5–50	++a 50–900	−I1r mPsb	58–70	>2000	200	183	16

TABLE 3. (*Continued*)

Site	% Oqapues	Color index[1]	Fe_3O_4	Fe_2O_3	Other	Blocking temperature in 10 °C	m.d.f.[2]	Intensity × 10^6 e.m.u.	k	F
N2 lavas										
W	2.3	A–E	++scp 60–200	+a 2–40	—	—	150	1230	26	3.6
S	4.5	E	++Bcp 40–200	+cp 40–200	—	Dist.	180	1000	62	2.4
V	3.2	B–E	++s 2–100	−Ba 100–500	—	54–70	190	1000	96	1.7
Q	1.7	A–E	++d 10–100	+a 50–500	—	Dist. and 58–62	180	1030	75	10
I	1.9	E	—	++Bs 50–150	—	Dist. and 58–62	320	8900	98	19
BB*	1.1	C–E	++a 1–20	−a 10–100	=Cu	Dist. and 30–40	>2000	22.3	44	4.0
R2 lavas										
U	5.5	D–F	—	++Ba 10–50 & 100–400	—	Dist. 50–70	>1000	140	188	31
R	3.0	B–E	−l 1–100	++Ba 1–100 & 150–300	−I11 5–100	58–62	>1000	311	32	2.8
J	3.3	A	—	++Bac 2–10 & 30–300	—	66–70	>2000	135	633	16
N	2.7	D–F	—	++Ba 100–300 & g2–500	−I11 10–60	54–66	200	84	5.4	73
A	5.1	B–C	++ap 5–150	+a 100–300	=I11tPsb	—	460	940	17	43
H	3.9	D–E	—	++Ba 200–800 & S1–200	−Psb	50–70	>2000	270	13	12
T	2.2	E	−mrl 5–50	++Bzl 5–300, zf 1–50	−I11 5–80	58–70	>1000	720	1580	3.4
K	1.9	A	++Bsc 60–600, a 1–30	—	=Cua	—	175	1700	10	73
Z*	3.4	F	−r 1–50	++Bd 2–60, z10–250	—	Dist. and 58–66	480	11500	84	3.4
AA*	3.5	B–E	++Bjd 5–400s 20–80	—	=Mb tCu	Dist. and 58–62	950	1140	658	4.4

[1]Wilson and Haggerty (1966).
[2]Median destructive alternating field in Oersteds (Park and Irving 1970).
The site letters refer to the sites of Fig. 1 and Table 2. Fe_3O_4 = magnetite, Fe_2O_3 = hematite, Other = other opaque minerals in the section. Il = ilmenite, Cu = native copper, Ps = pseudobrookite, Mh = maghemite, Su = unidentified sulfides. Dist. signifies distributed blocking temperature. k is the site precision after cleaning in 100 or 200 Oe. F = the ratio of the intensity of the most strongly to the most weakly magnetized specimen at the site: it is a measure of within-site intensity inhomogeneity. In the mineral descriptions the + and − denote the amount of the mineral: ++ = more than 50% of total opaques, + = 25 – 50%, − = 5–25%, = = less than 5%; letters describe the shape as follows: B = bimodal, s = subhedral, a = anhedral, f = fretted, r = relict, l = lamellar, d = dendritic, g = graphic intergrowth, c = corroded, p = poikalitic, b = blades, j = rod-like, z = alteration product. Numbers refer to the size range in microns.
[3]Site F was excluded from the paleomagnetic statistics.
*Denotes Gargantua Head sites.

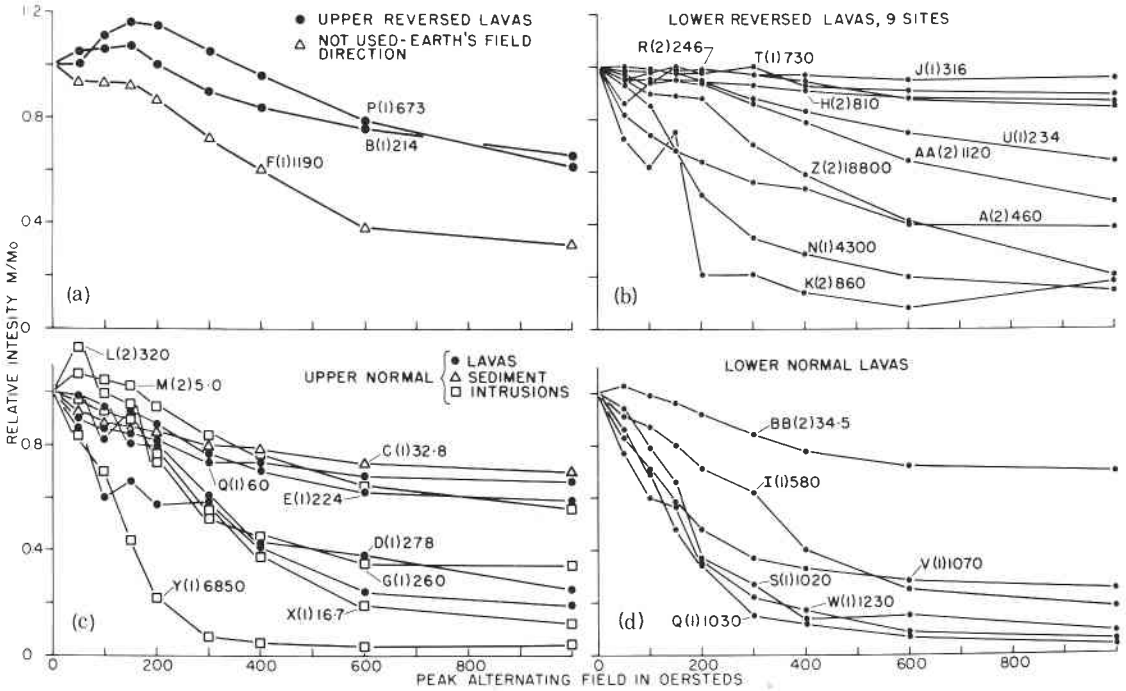


FIG. 3. Normalized alternating field demagnetization curves from (1) or (2) specimens from each site. Letters refer to the sites of Fig. 1 and the numbers to the initial intensity of magnetization in e.m.u. $\times 10^{-6}$.

initial cooling, possibly as oxidation during deuteric alteration, rather than as a later burial or weathering effect.

One or Three Reversals

The polarity succession in the Mamainse flows, from the base upwards is R2 → N2 → R1 → N1. This may be explained by three reversals in the earth's magnetic field at the times represented by the boundaries between the polarity changes. On the other hand, it can also be explained by a single polarity change at the time of the formation of the N1 – R1, N2 – R2 boundary, which has been repeated by strike faulting.

A single polarity change from older reversed to younger normal has been described from other areas in the Keweenawan. Palmer (1970) showed this relationship to be true of the North Shore volcanics, and also that the lower part of the Osler volcanics were reversely magnetized, and Halls (1972) showed from aeromagnetic anomalies that the upper part of these flows, mainly under Lake Superior, is normally magnetized. Also DuBois (1962) and Robertson and Fahrig (1971) have shown that the reversely magnetized Logan sills are older than the normally magnetized Thunder Bay dikes. At Gargantua Head a single polarity change from reversed to normal has been obtained by Palmer (1970) and in the present work.

In each case the poles from the normal group plot more than 30° W of those for the reversed group, each group being distinct and separate (see Fig. 5). Hence, if this difference in pole position is the result of movement of the region relative to the pole, then the poles from a continuous time sequence should lie along the great circle joining the two end positions. The Mamainse lava flows are such a serial time sequence. Hence, if there is no repetition by faulting the N2 directions should lie between those for R1 and R2, and the R1 directions between N1 and N2. In Fig. 4 the mean directions for each of the groups is plotted with the 95% error ovals calculated using sites as independent units. Unfortunately,

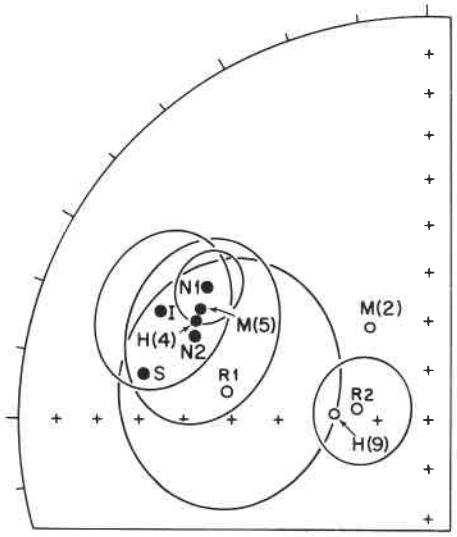


FIG. 4. An equal-area stereographic net segment showing the mean magnetization directions for the N1, and N2 flows and the intrusive group (I) with their error ovals, also the magnetization direction for the sediment site (S). The inverted (as though they were normal) mean directions of the two reversed groups, R1 and R2 with their error ovals are shown as open circles for comparison. The mean directions of the magnetite rich (M) and hematite rich (H) sites are also given, the numbers in brackets denote the number of sites.

the oval for R1 is large, as it was only possible to obtain directions from two flows in this thin group, but it is evident that R1 does not lie between N1 and N2, nor does N2 lie between R1 and R2. Also the N1, N2, and intrusive groups are themselves well grouped and significantly different from the R2 direction. Although the magnetic evidence is not entirely conclusive, it is strongly biased in favor of the hypothesis of repetition by faulting of a single reversal of the earth's field from older reversed polarity to younger normal polarity.

The Keweenawan Reversal

If the sequence (N1, R1, N2, R2) at Mamainse Point is accepted as the repetition of rocks formed during a single polarity change of the earth's magnetic field about 1000 m.y. ago, then the sequence of magnetic events in Keweenawan rocks is consistent. As suggested by Palmer (1970) a period in which the earth's magnetic field was reversed was followed by one in which it was normal. Since a considerable time of single polarity seems to have occurred both above and below this reversal it might be possible to call it the 'Middle Keweenawan Reversal' without ambiguity, and as such may provide a valuable worldwide chronostratigraphic marker horizon.

A most interesting aspect of this Middle Keweenawan Reversal is that wherever it has been found the pole position from the older, reversely magnetized rocks plots 30° or more west of that for the younger, normally magnetized rocks. Palmer (1970) suggested that this was because a secondary component common to both rock groups had not been eliminated by the cleaning, and he cited a conglomerate test as evidence in support of this idea. The de-

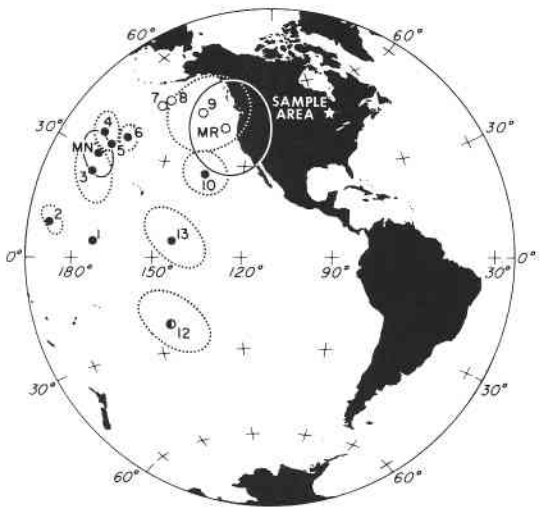


FIG. 5. An equatorial equal-area projection of the New World hemisphere of the earth (160° E–20° W) showing Keweenawan pole positions relative to North America, with P = 0.95 confidence ovals for the better controlled poles. 1 = Mackenzie (Fahrig and Jones 1969); 2 = Freda and Nonesuch (DuBois 1962); 3 = Portage Lake lavas (Vincenz 1968); 4 = Thunder Bay dikes (Robertson and Fahrig 1971); 5 = North Shore lavas normal (Palmer 1970); 6 = Duluth Gabbro (and associated intrusive rocks) (Beck 1970); MN = Mamainse Point lavas normal (this paper); 7 = North Shore lavas reversed (Palmer 1970); 8 = Osler lavas (Palmer 1970); 9 = Logan sills (Robertson and Fahrig 1971); MR= Mamainse Point lavas reversed (this paper); 10= Abitibi Dykes (Larochelle 1966); 12 = Sibley sediments (Robertson 1973); 13 = Croker Island Complex (Palmer 1969). Poles from normally magnetized rocks are plotted as solid circles and from reversely magnetized rocks as open circles; mixed polarity is denoted by a half-moon.

TABLE 4. Conglomerate tests

Formation	D	I	N	R	Ro	k	α
Portage Lake – Copper Harbour	311	+09	37	12.8	9.8	1.5	33
Mamainse Point	284	+35	37	11.5	9.9	1.4	36

D is the mean declination east of true north. I is the mean inclination positive downwards. N is the number of samples (separate boulders). R is the vector sum of the unit vectors from these samples. Ro is the value of R below which the direction may be considered random at $P = 0.95$ for this number of samples (Watson 1956). k is Fisher's (1953) best estimate of precision. α is the half angle of the cone of confidence at $P = 0.95$.

tailed laboratory work does not confirm this, however, and different rock types and groups of the same polarity plot close together with no sign of streaking (see Figs. 4 and 5).[4]

The good grouping (Fig. 5) of the normally magnetized parts of formations, such as the North Shore volcanics, Duluth Gabbro, Thunder Bay dikes, the Mamainse volcanics, and the Michipokoten volcanics, rocks that have different attitudes and hence require different rotations to correct for bedding attitude, is hard to reconcile with a direction made up of two components. Even the larger scatter of the reversely magnetized groups (7, 8, 9, and MR of Fig. 5) is too small to accommodate the difference between their poles and those of the normal groups. However, if the difference between the pole positions of normally and reversely magnetized rocks is caused by a change in the direction as well as the sense of the primary field, some other explanation is necessary for the directed component in the conglomerates.

Conglomerate Test

The conglomerate test, as devised by Graham (1949), has been used to show that the magnetization direction from boulders engulfed in a conglomerate predate its formation. If the directions of magnetization in boulders of an older formation are random, then these directions must have been acquired before the conglomerate was formed. Palmer (1970) used the inverse of this conglomerate test to say that since the directions in the conglomerate were not random, a secondary component must have been acquired since it was formed. Table 4 gives the Fisher (1953) statistics of the two conglomerates Palmer used as examples. It shows that in both cases, although the scatter is very high, a fact used by DuBois (1962) to indicate considerable magnetic stability of the parent lavas, the directions are not random at the $P = 0.95$ level, so that they do at first glance constitute a negative test of stability.

If we look at the situation closely we see that the boulders may contain a small, uniformly directed secondary component superposed on a primary component that is randomly oriented from boulder to boulder as shown in Fig. 6. Three factors may individually or jointly impress such a preferred direction on the magnetization directions in the boulders of the conglomerate. Firstly, the shape of the boulders may be elongated along the planes of the lava flows (or in some other direction) so that there is a tendency for them to settle in the conglomerate in similar attitudes. Secondly, some of the boulders may have been partly weathered after the conglomerate consolidated, and have acquired a secondary magnetization as a result of this weathering (*e.g.* Robertson and Hastie 1962). It must be remembered that only a small proportion of the total magnetization needs to be acquired in this way to yield the statistics of Table 4. Thirdly, in some instances the boulders may have been heated.

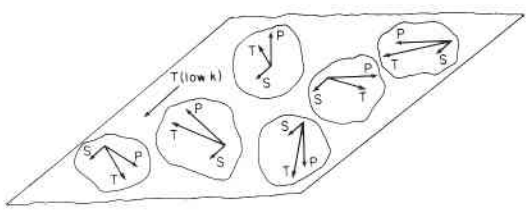


FIG. 6. Diagrammatic representation of a conglomerate in which the boulders contain a small directed secondary component (S) superposed on a large, randomly oriented primary component (P), yielding a widely scattered group with a mean in the direction of S.

[4]The exception, group R1, is probably the result of scanty sampling; possibly site B has suffered some undetected slumping.

This latter explanation is considered to be likely in the case of the Mamainse Point test.

The samples used in Palmer's conglomerate test were taken from two horizons. Seventeen samples were taken from a shoreline exposure of conglomerate containing rounded boulders of basaltic lava, slightly upsection from sites X and Y. The remaining 20 samples were taken from a roadcut in a most unusual rock, best described as a basalt breccia (Annells 1973). It contains angular to subangular fragments of medium-grained basalt in a fine-grained basaltic matrix. Some of the fragments are amoeboid in shape and the deposit is generally reminiscent of hyaloclastite breccias formed by eruptions in shallow water (Annells 1973). If so, many of the fragments can be expected to have come to rest at elevated temperatures and to have cooled, at least partly, in the ambient field. Therefore, such fragments would almost certainly have acquired a component in the direction of the field at that time.

These considerations lead to the conclusion that the negative conglomerate test provides doubtful evidence for secondary components retained in the samples after magnetic cleaning. Furthermore, analysis of Palmer's argument (1970, pp. 1430–31) in favor of a retained secondary component reveals a paradox. In his Fig. 22 he cited two possible cases. (*1*) In the first of these (Fig. 22A), the initial normal and reversed magnetizations have the same intensity. This is a plausible hypothesis, but it yields normally magnetized lavas with higher intensities than those reversely magnetized, which is contrary to the measured results, in which the mean intensity of the normal and the reverse flows is almost identical (Palmer 1970, p. 1431). (*2*) His other case (Fig. 22B) fits the measured intensity requirements, but requires that the initial magnetization of the reversed lavas is more than twice as strong as that of the lavas normally magnetized. Although the intensity of the earth's magnetic field is known to oscillate (Smith 1967), it does not seem likely that the mean field intensity during the outpourings of the R lavas was double that at the time the N lavas were extruded, nor is there any petrological evidence (Table 3) to indicate that the R lavas might become more strongly magnetized.

If the difference between the pole positions derived from normally and reversely magnetized lavas is the result of incomplete removal of a secondary component after cleaning, a major problem must be resolved. What constitutes a stable magnetization direction? The Mamainse Point lavas (*see* Table 1) have been subjected to progressive thermal and alternating field demagnetization, and yet the difference in pole position persists until the directions become random. If they retain a secondary component, why should not other results from rocks treated similarly or more leniently also retain one? In that case only results that are confirmed by a well-defined stability test (such as a fold, conglomerate, or slump formed soon after the rock consolidated, or a baked contact (Graham 1949; Irving 1964)) are suitable to yield meaningful pole positions, and most points now used to define polar wandering paths would have to be discarded as unreliable.

Polar Shift

If the difference in pole position between the normally and reversely magnetized lavas is not caused by a secondary component, then some other explanation is necessary. One obvious possibility is that the earth's magnetic field was not that of a geocentric axial dipole at the time. This is hard to test, but the considerable body of consistent data from formations of this age and older that is based on the geocentric axial dipole field hypothesis (*see* Hicken *et al.* 1972) provides a good foundation for accepting the hypothesis in broad outline, although allowing for deviations from it in detail (*e.g.* Wilson 1970).

Pole positions from the younger normally magnetized lavas lie closer to those from younger formations such as the Frieda and Nonesuch (DuBois 1962) and Franklin (Fahrig *et al.* 1971) than do the poles from the older reversely magnetized lavas, which lie closer to that of the older Abitibi dikes (Fig. 5). It seems natural, therefore, to follow DuBois (1962) and Robertson and Fahrig (1971) in postulating that the difference between the normally and reversely magnetized groups of poles (Fig. 5) is the result of relative movement of the Canadian Shield and the pole,

that is to say, of polar wandering or continental drift.[5]

The rate of apparent polar movement between the position of the older (R) poles and the younger (N) poles is not known because of uncertainties in the dating of the rocks from which they were derived (Robertson and Fahrig 1971). However the most likely dates indicate higher rates of movement (>20 cm/yr) then those associated with Tertiary ridge spreading (Heirtzler *et al.* 1968). Hence the polar shift indicated between the times of formation of the reversely (R) and normally (N) magnetized rock may represent a period of polar wandering in which possibly the crust as a whole may have slipped over the asthenosphere relative to the rotational axis, comparable with that at the time of the Patterson transition (Irving and Robertson 1969).

Several consequences follow from postulating such a movement. Firstly a movement of over 30° of arc, a distance on the surface of the earth of over 3000 km (>1900 mi), must have taken considerable time. Assuming the very fast rate of 30 cm/yr (12 in./yr) the time involved is over 10 m.y. The passage of such a length of time should be apparent on the ground as a disconformity, and might even be large enough to be evident in small radiogenic age differences between the normally and reversely magnetized rocks. Secondly, movement must have been spasmodic (Briden 1967). The pole and continent remained fixed during the extrusion of the whole mass of normal lavas, and also earlier during the extrusion of the reverse lavas, whereas movement during the interval between the two must have been rapid. Thirdly, the low paleolatitude of the normally magnetized rocks (15–25°) should yield tropical weathering, such as the formation of red beds as seen in the conglomerates, whereas the reversely magnetized groups have a high paleolatitude (45–70°). A difference in the type of weathering as a result of such a latitude change (at least 20°) should be apparent not only in the Mamainse Point rocks, but also in the North Shore volcanics, and anywhere that has been subjected to this polar jump.

Failure to obtain a positive test from any of these lines of evidence does not necessarily mean that the hypothesis of a polar shift is wrong, but it would encourage the search for other hypotheses that fit the known facts. A positive test confirming the difference of paleolatitude, on the other hand, would be strong support for the idea that the apparent movement of the pole between the time of formation of the reversely and normally magnetized rocks is the result of movement of this part of the shield (and perhaps the whole crust) relative to the pole of the earth's magnetic field.

[5]It has been pointed out (W. R. A. Baragar and R. N. Annells, personal communication) that the great conglomerate, which marks the base of the N1 succession, indicates a time of great crustal subsidence during which the extrusion of lavas ceased. This period of crustal instability may have extended all round Lake Superior, and it is possible that it may have caused tilting of the lower beds. It does not seem possible, however, that such tilting caused the difference between the pole positions of the normal and the reversed groups, although it may have contributed towards the greater scatter evident in the reversely magnetized formations.

Acknowledgments

It is a pleasure to acknowledge the help and advice given during the field collecting by Jean Roy, Dick Annalls, and Bob Baragar. I also wish to thank Ric Lavery for making many of the measurements. Finally I thank Bob Baragar and Ted Irving for critically examining the manuscript and offering suggestions for its improvement.

ANNELLS, R. N. 1973. Proterozoic flood basalts of eastern Lake Superior: the Keweenawan volcanic rocks of Mamainse Point area, Ontario. Geol. Surv. Can., Pap. 72-10.

AYRES, L. D. 1961. Township 31, Range 19. Ont. Dept. Mines, Prel. Geol. Map No. P.128.

BECK, M. E. 1970. Paleomagnetism of Keweenawan intrusive rocks, Minnesota. J. Geoph. Res., **75**, pp. 4985–4996.

BOOKS, K. G. 1968. Magnetization of the lowermost Keweenawan lava flows in the Lake Superior Area. U.S. Geol. Surv. Prof. Pap. **600D**, pp. 248–254.

BRIDEN, J. C. 1967. Recurrent continental drift of Gondwanaland. Nature, **215**, pp. 1334–1339.

DUBOIS, P. M. 1962. Palaeomagnetism and correlation of Keweenawan rocks. Geol. Surv. Can., Bull. 71, 75 p.

FAHRIG, W. F., IRVING, E., and JACKSON, G. D. 1971. Paleomagnetism of the Franklin diabases. Can. J. Earth Sci., **8**, pp. 455–467.

FAHRIG, W. F. and JONES, D. L. 1969. Paleomagnetic evidence for the extent of Mackenzie igneous events. Can. J. Earth Sci., **6**, pp. 679–688.

FAURE, G., CHAUDHURI, S., and FENTON, M. D. 1969. Ages of the Duluth Gabbro complex and of the Endion sill, Duluth, Minnesota. J. Geophys. Res., **74**, pp. 720–725.

FISHER, R. A. 1953. Dispersion on a sphere. Proc. Roy. Soc. London, A, **217**, pp. 295–305.

GIBLIN, P. E. 1969. Kinaird Township. Ont. Dept. Mines Prelim. Geol. Map No. P553.

GIBLIN, P. E. and LEAHY, E. J. 1965. Batchawana sheet. Ont. Dept. Mines, Prel. Geol. Map. No. P.302.

GRAHAM, J. W. 1949. The stability and significance of magnetism in sedimentary rocks. J. Geophys. Res., **54**, pp. 131–167.

HALLS, H. C. 1972. Magnetic studies in northern Lake Superior. Can. J. Earth Sci., **9**, pp. 1349–1367.

HEIRTZLER, J. R., DICKSON, G. O., HERRON, E. M., PITMAN, W. C., and LE PICHON, X. 1968. Marine magnetic anomalies, geomagnetic field reversals, and motions in the ocean floor and continents. J. Geoph. Res., **73**, pp. 2119–2136.

HICKEN, E., IRVING, E., LAW, L. K., and HASTIE, J. 1972. Catalogue of paleomagnetic directions and poles. Can. Dept. Energy, Mines and Resources, Publ. Earth Phys. Br., **45**, pp. 1–135.

IRVING, E. 1964. Paleomagnetism and its application to geological and geophysical problems. John Wiley and Sons, New York, 399 p.

IRVING, E. and ROBERTSON, W. A. 1969. Test for polar wandering and some possible implications. J. Geoph. Res., **74**, pp. 1026–1036.

LAROCHELLE, A. 1966. Paleomagnetism of the Abitibi dyke swarm. Can. J. Earth Sci., **3**, pp. 671–683.

MCELHINNY, M. W. 1973. Palaeomagnetism and plate tectonics. Cambridge Univ. Press, London, 358 p.

NEEL, L. 1955. Some theoretical aspects of rock magnetism. Phil. Mag. Supp., **4**, pp. 191–243.

PALMER, H. C. 1969. The paleomagnetism of the Croker Island Complex, Ontario, Canada. Can. J. Earth Sci., **6**, pp. 213–218.

——— 1970. Paleomagnetism and correlation of some Middle Keweenawan rocks, Lake Superior. Can. J. Earth Sci., **7**, pp. 1410–1436.

PARK, J. K. and IRVING, E. 1970. The Mid-Atlantic Ridge near 45° N. XII. Coercivity, seconding magnetization, polarity, and thermal stability of dredge samples. Can. J. Earth Sci., **7**, pp. 1499–1514.

ROBERTSON, W. A. 1973. Pole position from thermally cleaned Sibley Group sediments and its relevance to Proterozoic magnetic stratigraphy. Can. J. Earth Sci., **10**, pp. 180–193.

ROBERTSON, W. A. and FAHRIG, W. F. 1971. The Great Logan paleomagnetic loop—the polar wandering path from Canadian Shield rocks during the Neohelikian Era. Can. J. Earth Sci., **8**, pp. 1355–71.

ROBERTSON, W. A. and HASTIE, L. 1962. A paleomagnetic study of the Cygnet Alkaline Complex of Tasmania. J. Geol. Soc. Aust., **8**, pp. 259–268.

ROY, J. L., REYNOLDS, J., and SANDERS, E. 1971*a*. An alternating field demagnetizer for rock magnetic studies. Can. Dept. Energy, Mines and Resources, Publ. Earth Phys. Br., **42**, No. 5.

ROY, J. L., SANDERS, E., and REYNOLDS, J. 1971*b*. Un four éléctrique pour l'étude des propriétés magnétiques des roches. Can. Dept. Energy, Mines and Resources, Publ. Earth Phys. Br., **42**, No. 8.

SCHWARZBACH, M. 1963. Climates of the past. Van Nostrand, New York, 328 p.

SILVER, L. T. and GREEN, J. C. 1963. Zircon ages for Middle Keweenawan rocks of the Lake Superior region (Abstr.). Trans Am. Geophys. Union, **44**, 107 p.

SMITH, P. J. 1967. The intensity of the ancient geomagnetic field: A review and analysis. Geophys. J. Roy. Astron. Soc., **12**, pp. 321–362.

THOMSON, J. E. 1953. Geology of the Mamainse Point copper area. Ont. Dept. Mines, Ann. Rep., **62**, 25 p.

VAN HOUTEN, F. B. 1961. Climatic significance of red beds; *In* Descriptive palaeoclimatology (A. E. M. Nairn *Ed.*). Interscience, New York, pp. 89–139.

VAN SCHMUS, W. R. 1971. Rb–Sr age of Middle Keweenawan rocks, Mamainse Point and vicinity, Ontario, Canada. Bull. Geol. Soc. Am., **82** pp. 3221–3226.

VINCENZ, S. A. 1968. Phenomenon of partial self-reversal in Keweenawan rocks. 1. Magnetization of Portage Lake Lavas. J. Geoph. Res., **73**, pp. 2729–2752.

WANLESS, R. K., STEVENS, R. D., LACHANCE, G. R., and EDMONDS, C. M. 1967. Age determinations and geological studies, K–Ar isotopic ages. Rep. 7. Geol. Surv. Can., Pap. 66-17, 120 p.

——— 1968. Age determinations and geological studies, K–Ar isotopic ages. Rep. 8. Geol. Surv. Can. Pap. 67-2, Part A, 141 p.

WATSON, G. S. 1956. A test for randomness of directions. Mon. Not. Roy. Astron. Soc., Geophys. Supp., **7**, pp. 160–161.

WHITE, W. S. 1960. The Keweenawan lavas of Lake Superior, an example of flood basalts. Am. J. Sci., **258** A (Bradley Vol.), pp. 367–374.

WILSON, R. L. 1970. Permanent aspects of the earth's non-dipole magnetic field over Upper Tertiary times. Geophys. J. Roy. Astron. Soc., **19**, pp. 417–437.

WILSON, R. L. and HAGGERTY, S. E. 1966. Reversals of the earth's magnetic field. Endeavour, **15**, pp. 104–109.